

Municipality_Pemex_Distances Data Dictionary

All distances in kilometers, using EPSG:6372 (Mexico ITRF2008 / LCC)

I. Naming Convention

Most column names are built from 3 parts: <measure>_<facilitytype>_<reference>

<measure>

Prefix	Meaning
dist	Distance in km to the nearest facility of that type
nearest	Name/ID of the specific facility that was nearest
n	Count of facilities of that type within a radius band

<facilitytype>

Suffix	Facility Type
refinery	Refinery
gasproc	Gas processing complex (CPG)
petrochem	Petrochemical complex
storage	Storage & distribution terminal (TAR)
maritime	Marine terminal
oilfield	Oil field
any	All six types pooled

<reference>

Suffix	Meaning
_cent	The municipality's centroid (interior point). For 66 municipalities whose true centroid fell outside the polygon, a point-on-surface was substituted.
_edge	The municipality's boundary. Shortest distance from any part of the polygon. Equals 0 when the facility is inside the municipality.

II. Distance Columns:

Column	Description
dist_refinery_cent	km from centroid to nearest refinery
dist_refinery_edge	km from boundary to nearest refinery
dist_gasproc_cent	km from centroid to nearest gas processing complex
dist_gasproc_edge	km from boundary to nearest gas processing complex
dist_petrochem_cent	km from centroid to nearest petrochemical complex

dist_petrochem_edge	km from boundary to nearest petrochemical complex
dist_storage_cent	km from centroid to nearest storage/distribution terminal
dist_storage_edge	km from boundary to nearest storage/distribution terminal
dist_maritime_cent	km from centroid to nearest marine terminal
dist_maritime_edge	km from boundary to nearest marine terminal
dist_oilfield_cent	km from centroid to nearest oil field
dist_oilfield_edge	km from boundary to nearest oil field
dist_any_cent	km from centroid to nearest facility of any type
dist_any_edge	km from boundary to nearest facility of any type

III. Interpretation Notes

Choosing _cent vs _edge. Centroid distance approximates distance for a typical location within the municipality; boundary distance measures pure adjacency and exposure. They diverge most in the large northern municipalities (Sonora, Chihuahua, Baja California).

_edge is censored at zero. Municipalities hosting a facility all get 0, so the variable is not continuous across its range. Consider pairing it with hosts_any rather than relying on the continuous measure alone.

Oil fields are points, but real fields are not. dist_oilfield_* measures distance to a single representative point for what may be a production area spanning tens of km. This understates proximity for municipalities overlapping the field's actual extent. Treat as measurement error, or substitute CNH field polygons if available.

Marine terminal distances cross water. Coastal municipalities in Campeche, Tabasco, and Veracruz will show small dist_maritime_* values measured over open water, with no road connection implied. Defensible for port facilities; check any value under ~5 km to confirm it's a shore facility rather than an offshore platform.

All distances are Euclidean, not travel distance. Given the Sierra Madre ranges, straight-line and road distance diverge substantially in mountainous regions (especially Oaxaca, Guerrero, Chiapas). This measurement is weaker for economic outcomes, where a road-network or travel-time version is more defensible.

Northern border facilities. If the facility layer is Mexico-only, municipalities along the US border may show inflated distances because relevant cross-border infrastructure is absent from the data.